Atomic modelling using Scipion-chem

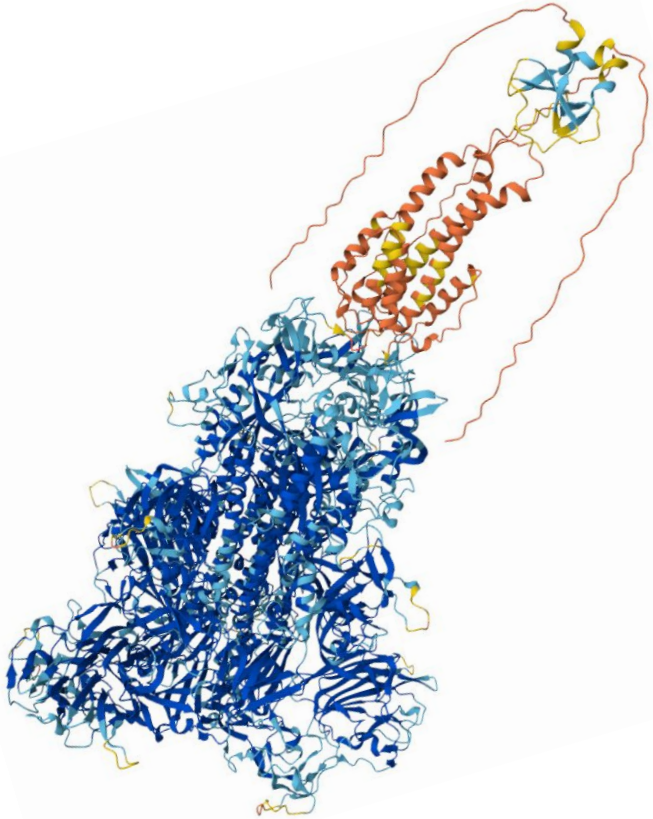
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Atomic modelling

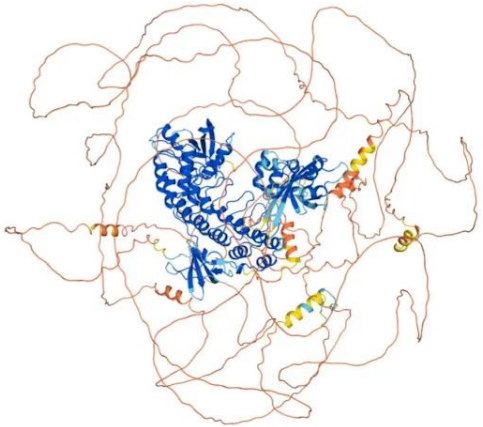
- Build **3D atomic models** of biomolecules
- Predict protein structures from **amino acid sequences**
- Model **biomolecular complexes**, including protein–protein, protein–DNA/RNA and protein–ligand interactions
- Refine existing models to improve their geometry and agreement with experimental observations

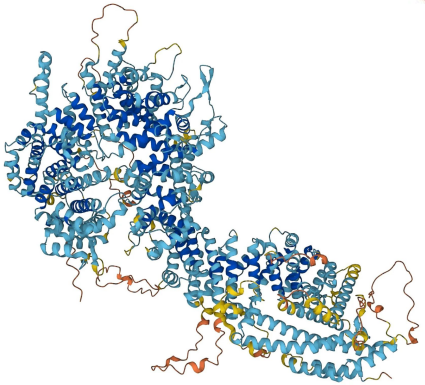
Main objective: *Obtain* ***reliable atomic structures*** *to investigate biomolecular structure, interactions and function*



BioFold: Integrated atomic modelling in Scipion

- **BioFold** is a Scipion plugin that provides access to different AI-based biomolecular structure-prediction tools
- Integrates several methods:
  - AlphaFold3
  - Protenix
  - IntelliFold
  - Boltz-2
  - Chai-1
- Allows users to **prepare** prediction inputs, **execute** supported methods and **retrieve** their outputs within Scipion
- Depending on the method and protocol, predictions can be **run locally** or **imported** directly from external servers
- Supports modelling tasks involving proteins and nucleic acids





How do these methods work?

- **Input preparation:** Provide the **molecular components** to be modelled, such as protein sequences, nucleic acid sequences or ligand descriptions
- **Information processing:** The model **learns relationships** between molecular components using information learned during training
- **Structure generation:** Deep-learning architectures **predict** the spatial arrangement of atoms and generate candidate 3D structures
- **Confidence estimation:** Models provide **confidence measures** to help assess the reliability of their predictions

- **Multiple sequence alignments (MSAs):** Provide **evolutionary information** by identifying related sequences and patterns of conservation
- **Templates:** Previously determined structures can provide structural information to guide predictions
- **Diffusion models:** Generate atomic coordinates through an **iterative process** that progressively transforms **noisy coordinates into a plausible structure**

Key idea: *Different methods* ***combine sequence, evolutionary, structural and chemical information*** *in different ways to predict atomic coordinates*



AlphaFold 3

A general model for biomolecular structure prediction

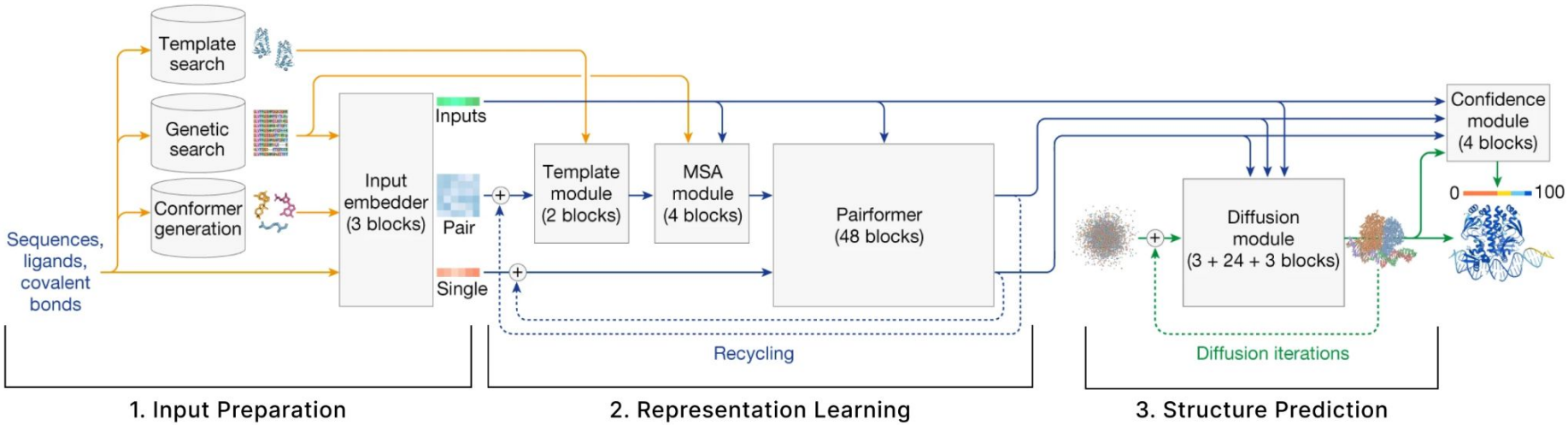
- Predicts structures of individual proteins and complexes containing different molecular components
- Supports proteins, DNA, RNA, small-molecule ligands, ions and modified residues
- Uses **evolutionary information** from sequence alignments, together with learned representations of molecular interactions
- Uses a **Pairformer** network to process molecular information and a **diffusion module** to generate atomic coordinates
- Provides confidence estimates to help assess predicted structures

What makes it important?

- Extends structure prediction beyond proteins to a broad range of biomolecular interactions
- Allows different molecular components to be modelled within a unified framework

Key idea: *Predicts the joint 3D structure of interacting biomolecules rather than treating each component independently.*

AlphaFold 3


Template search
Genetic search
Conformer generation
Sequences, ligands, covalent bonds
Input embedder (3 blocks)
Inputs
Pair
Single
Template module (2 blocks)
MSA module (4 blocks)
Pairformer (48 blocks)
Recycling
Diffusion module (3 + 24 + 3 blocks)
Diffusion iterations
Confidence module (4 blocks)
0
100
1. Input Preparation
2. Representation Learning
3. Structure Prediction

Protenix

An open-source implementation inspired by AlphaFold 3

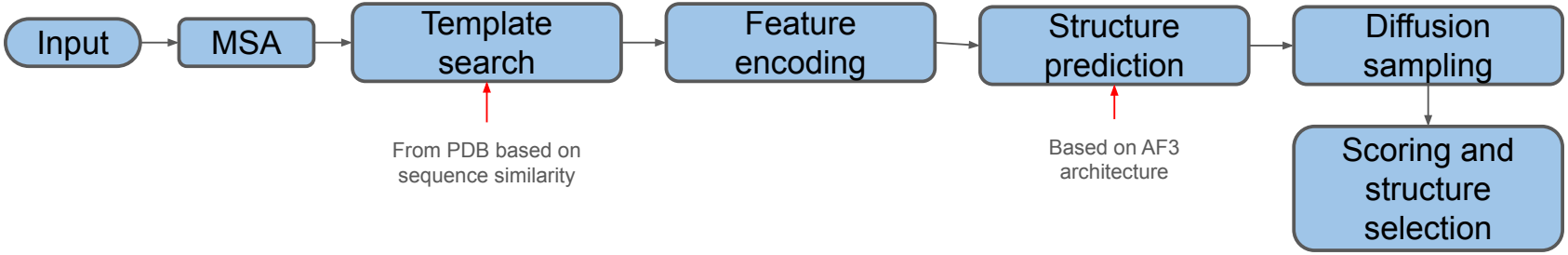
- Developed to reproduce and extend the biomolecular structure-prediction capabilities of AlphaFold 3
- Predicts structures involving proteins, ligands and nucleic acids
- Uses an **architecture based on the AlphaFold 3 framework**, including Pairformer-based processing and diffusion-based coordinate generation
- Supports evolutionary information through multiple sequence alignments
- Later versions also incorporate **additional features** such as **template information and RNA MSAs**

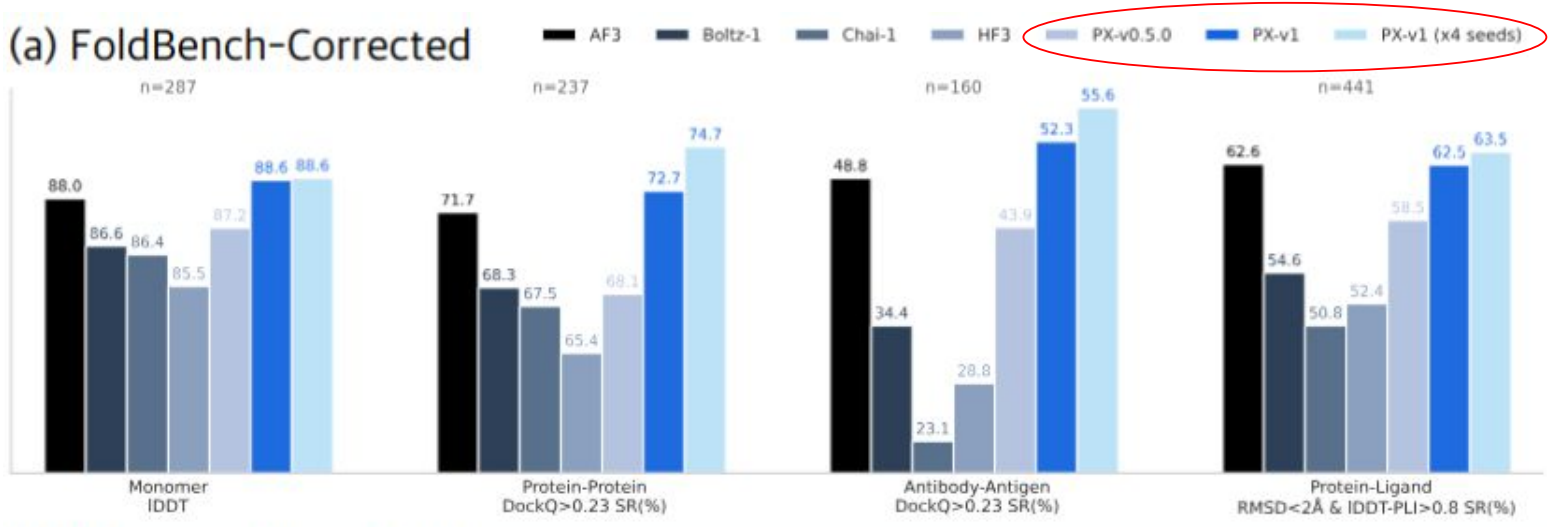
What makes it interesting?

- Provides an openly available alternative for running AF3-style structure prediction
- Allows local execution and further development of the prediction framework
- Increasing the number of prediction samples can improve the chance of obtaining accurate models, although this requires additional computational resources

Key idea: *Protenix makes AF3-style biomolecular structure prediction accessible through an open-source framework*

Protenix


Input
MSA
Template search
Feature encoding
Structure prediction
Diffusion sampling
Scoring and structure selection
From PDB based on sequence similarity
Based on AF3 architecture


(a) FoldBench-Corrected
AF3
Boltz-1
Chai-1
HF3
PX-v0.5.0
PX-v1
PX-v1 (x4 seeds)
n=287
n=237
n=160
n=441
88.0
86.6
86.4
85.5
87.2
88.6
88.6
71.7
68.3
67.5
65.4
68.1
72.7
74.7
48.8
34.4
23.1
28.8
43.9
52.3
55.6
62.6
54.6
50.8
52.4
58.5
62.5
63.5
Monomer IDDT
Protein-Protein DockQ>0.23 SR(%)
Antibody-Antigen DockQ>0.23 SR(%)
Protein-Ligand RMSD<2Å & lDDT-PLI>0.8 SR(%)

IntelliFold

An alternative deep-learning approach to biomolecular structure prediction

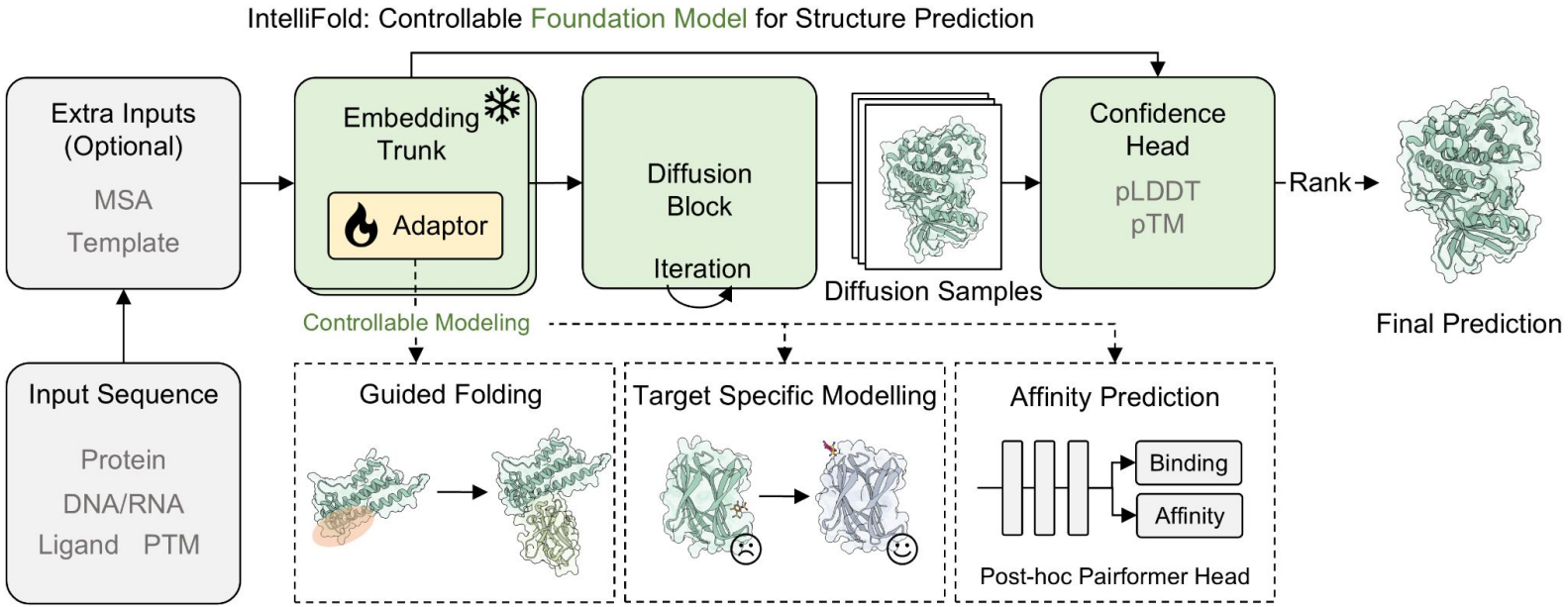
- Predicts biomolecular structures and complexes using an **AlphaFold 3-style framework**
- Uses learned representations of molecular components to capture structural and interaction information
- Generates candidate atomic structures using a **diffusion-based approach**
- Provides alternative predictions that can be compared with those from other structure-prediction methods

IntelliFold-2: architectural developments

- Introduces refinements to Pairformer-based processing
- Uses an improved atom-attention formulation
- Incorporates multiscale structural consistency during prediction
- Explores improved diffusion sampling strategies to enhance prediction quality

Key idea: *IntelliFold provides an alternative AF3-style prediction framework, with newer versions introducing architectural and sampling improvements*

IntelliFold



Boltz-2

Joint biomolecular structure and binding-affinity prediction

- Predicts 3D structures of biomolecular complexes, including protein–ligand interactions
- Uses an architecture based on Pairformer processing and diffusion-based coordinate generation
- Incorporates structural information to model interactions between molecular components

Additional capability: binding-affinity prediction

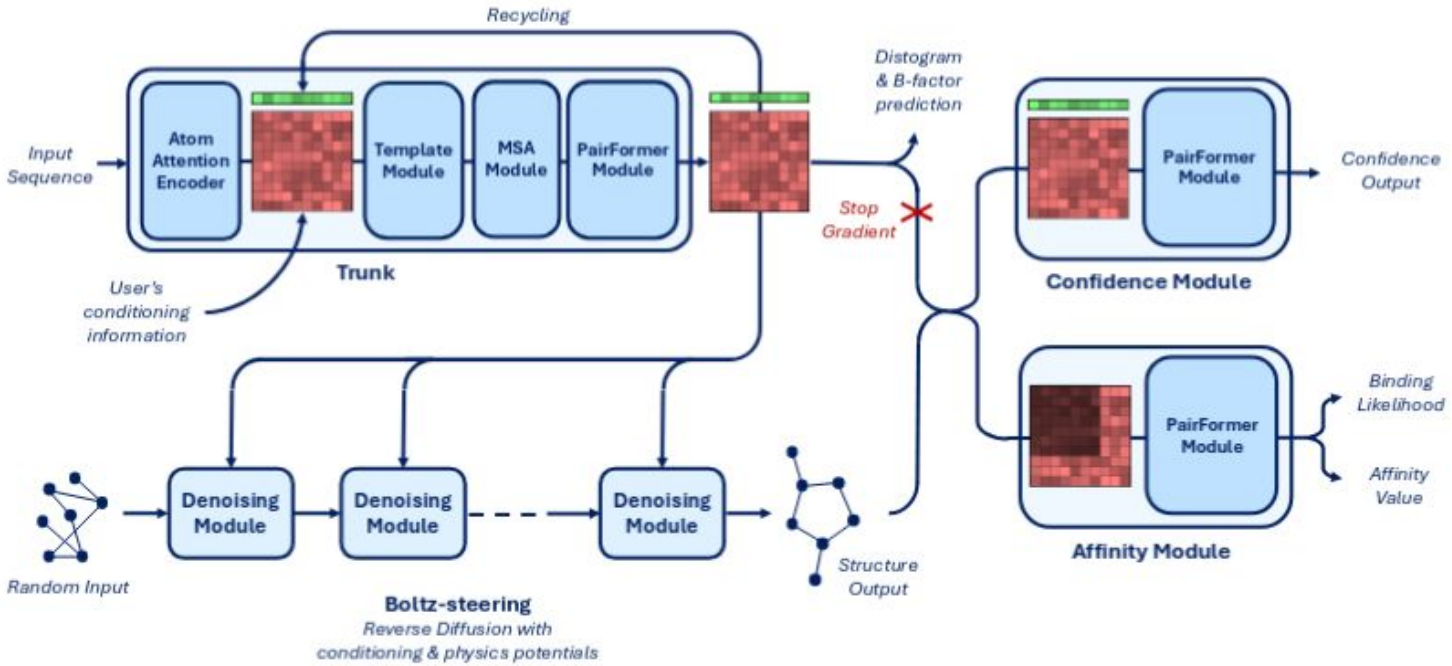
- Includes an **affinity module** to estimate protein–ligand **binding affinity**
- Designed to support molecular design and virtual screening applications

Why is this interesting?

- Combines structural prediction with an estimate of binding affinity
- Can **help prioritize compounds** for further investigation
- Provides complementary information to structure-based docking and other screening approaches

Key idea: *Boltz-2 goes beyond predicting where molecules may interact by also estimating the* ***strength of protein–ligand binding***

Boltz-2


Recycling
Input Sequence
Atom Attention Encoder
Template Module
MSA Module
PairFormer Module
Trunk
User's conditioning information
Distogram & B-factor prediction
Stop Gradient
PairFormer Module
Confidence Output
Confidence Module
PairFormer Module
Binding Likelihood
Affinity Value
Affinity Module
Random Input
Denoising Module
Denoising Module
Denoising Module
Structure Output
Boltz-steering
Reverse Diffusion with conditioning & physics potentials

Chai-1

A multimodal foundation model for molecular structure prediction

- Predicts structures involving proteins, DNA, RNA, small molecules and other molecular components
- Uses a deep-learning architecture **inspired by AlphaFold 3**
- Can also operate in single-sequence mode, without requiring MSAs

Additional capability: experimental restraints

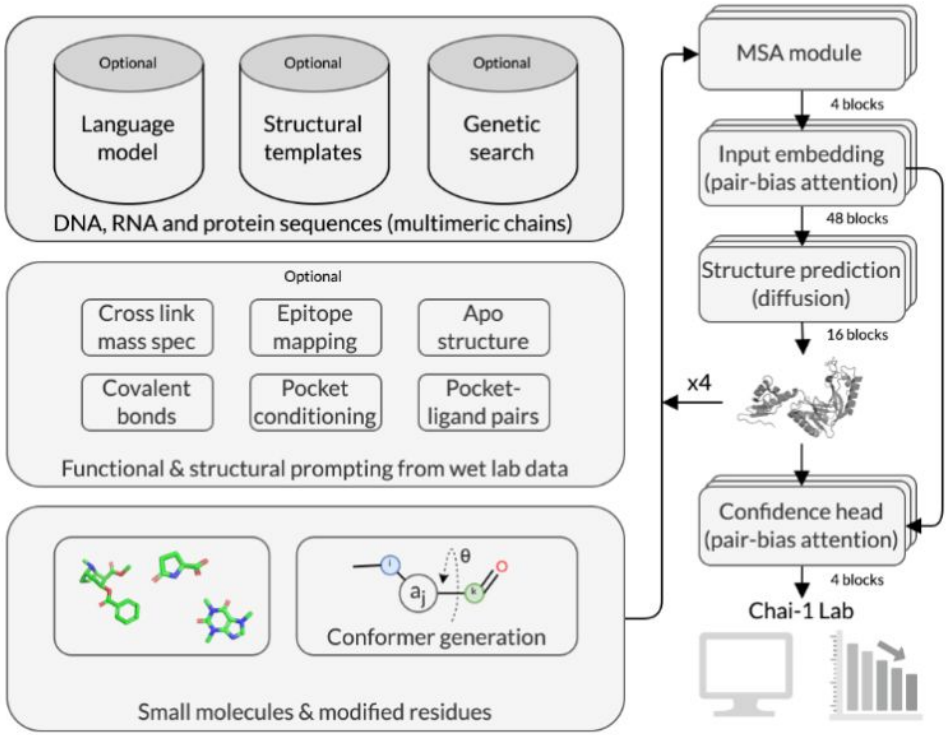
- Examples include restraints derived from cross-linking mass spectrometry and epitope mapping
- These constraints can help guide predictions of challenging biomolecular complexes

What makes it interesting?

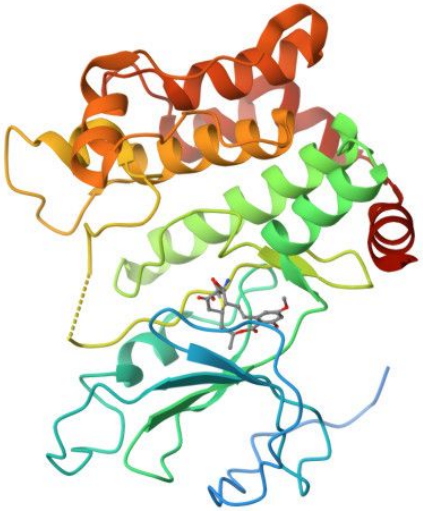
- **Combines sequence and chemical information** within a unified prediction framework

Key idea: *Chai-1 combines multimodal structure prediction with the possibility of incorporating* ***experimental constraints***

Chai-1


Optional
Language model
Optional
Structural templates
Optional
Genetic search
DNA, RNA and protein sequences (multimeric chains)
Optional
Cross link mass spec
Epitope mapping
Apo structure
Covalent bonds
Pocket conditioning
Pocket-ligand pairs
Functional & structural prompting from wet lab data
Conformer generation
Small molecules & modified residues
MSA module
4 blocks
Input embedding (pair-bias attention)
48 blocks
Structure prediction (diffusion)
16 blocks
x4
Confidence head (pair-bias attention)
4 blocks
Chai-1 Lab

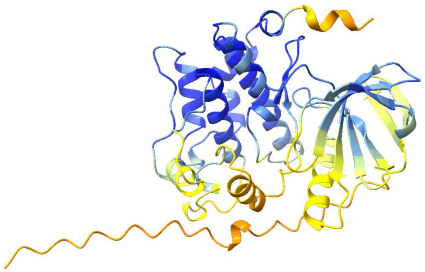
Use case: 9M1Z



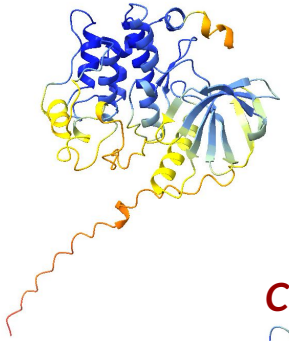
AlphaFold3



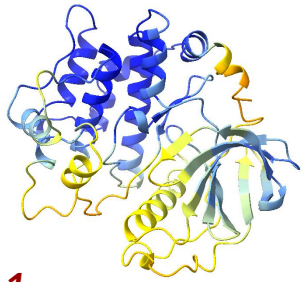
Protenix



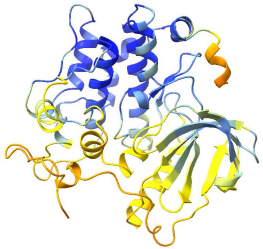
IntelliFold



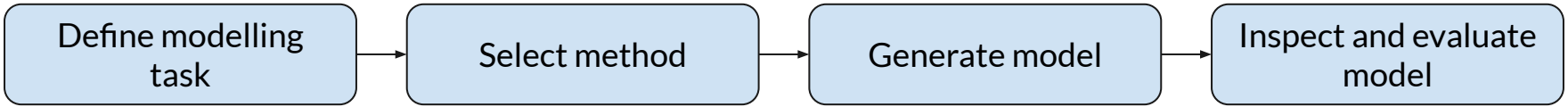
Boltz-2



Chai-1



Atomic modelling workflow



- Protein structure
- Protein complex
- *DNA or RNA?*

- Select **tool**
- *Run locally or run on server and import?*

- **Run** model or import

- Check structural **confidence**
- Examine **geometry** and **potential clashes**
- Asses whether the model is **consistent with available evidence**

Evaluating predicted atomic models

Confidence estimates

- **pLDDT:** Estimates **local confidence** in the predicted structure
- **PAE** (Predicted Aligned Error)**:** Estimates **uncertainty** in the relative positioning of different **residues** or **regions**
- **ipTM:** Provides information about the **confidence** of **predicted interfaces** in multimeric structures

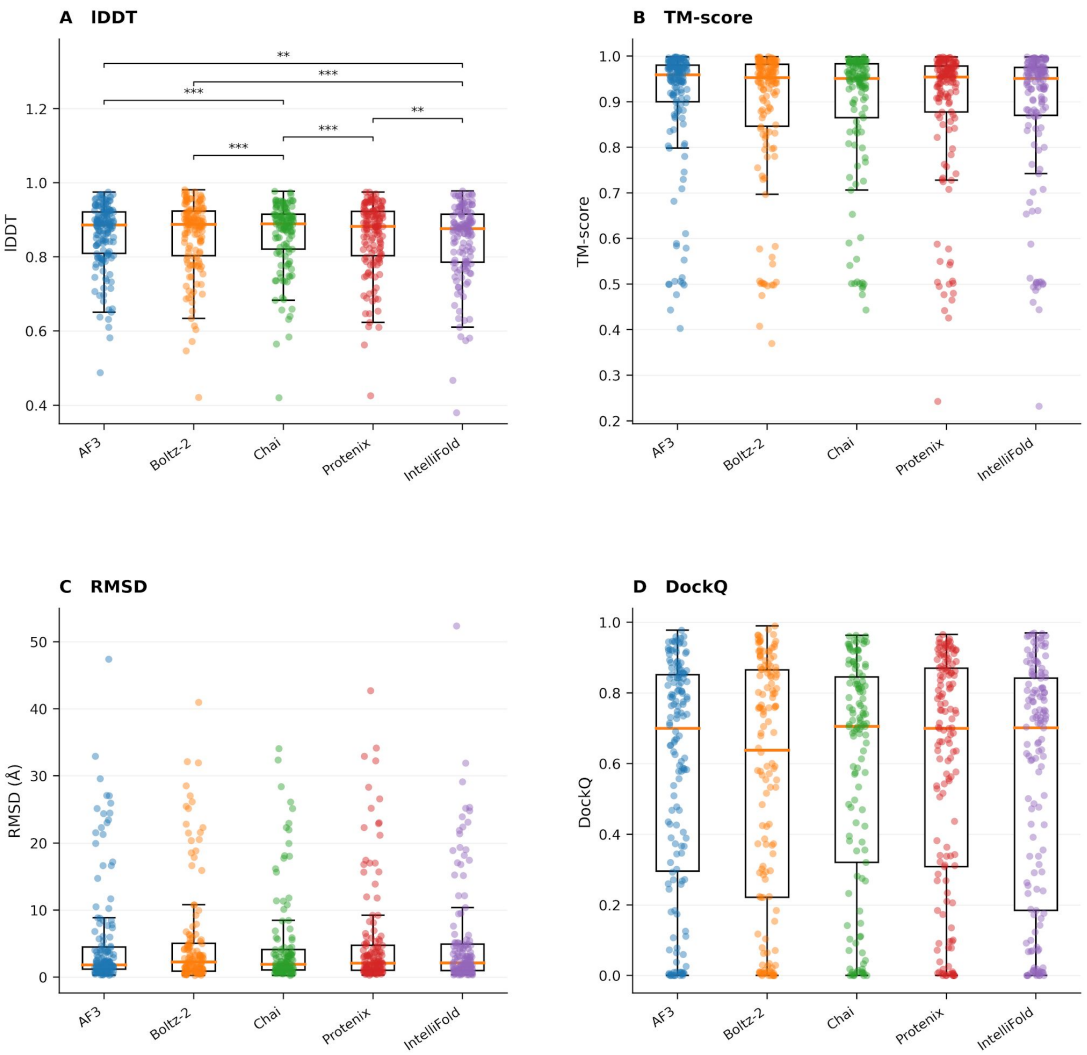
Structural quality

- **Steric clashes:** Identify atoms that are **unrealistically close** to one another
- **Bond lengths and angles:** Check whether **covalent geometry** is chemically reasonable
- **Stereochemistry:** Examine **chirality**, **torsion angles** and **other geometric properties**

Agreement with experimental evidence

- Compare predicted structures with available experimental models, structural data or assess agreement with the experimental density map
- Consider whether the predicted interactions and overall arrangement are biologically plausible

Key idea: *Confidence scores and geometric validation provide complementary info.* ***High-confidence*** *prediction* ***is not necessarily experimentally correct***


A IDDT
IDDT
**
***
***
***
***
**
AF3
Boltz-2
Chai
Protenix
IntelliFold
B TM-score
TM-score
C RMSD
RMSD (Å)
D DockQ
DockQ
* FDR-adjusted P < 0.05; ** FDR-adjusted P < 0.01; *** FDR-adjusted P < 0.001